

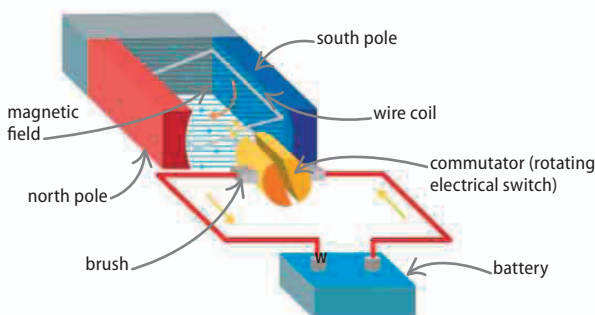
Electric motors

AN ELECTRIC CURRENT AND THE FORCES IN A MAGNETIC FIELD CAN COMBINE TO CREATE MOTION.

An electric motor turns because of the forces of attraction and repulsion between a permanent magnet and an electromagnet.

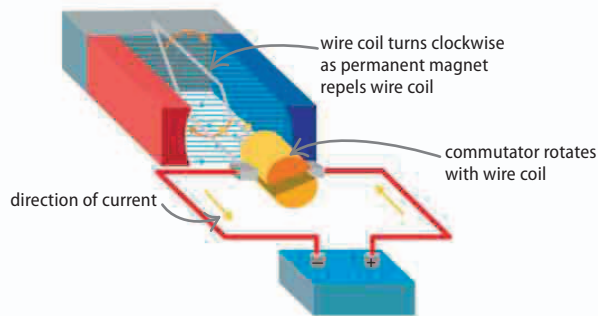
Inside a motor

A wire coil sits between the opposite poles of one or more permanent magnets. When an electric current is passed through the wire coil, it generates a magnetic field, which interacts with the magnetic field of the surrounding permanent magnets, repelling like poles and attracting unlike poles, which make the wire coil rotate half a turn. The electric current is then reversed to switch the wire coil's magnetic poles, so that it moves another half-turn. Repeating this process results in the coil spinning around.



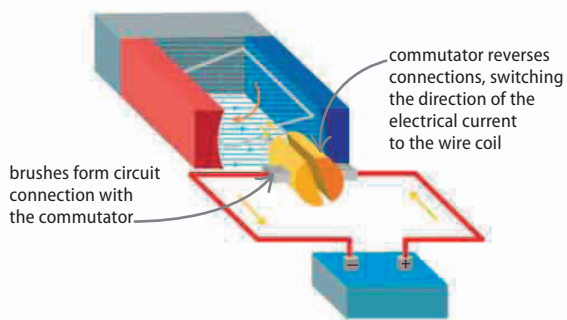
△ Stage 1

In this simple DC electric motor, current flows from the battery through the commutator and into the wire coil. This turns it into an electromagnet and generates a magnetic field, which interacts with the field of the permanent magnet.



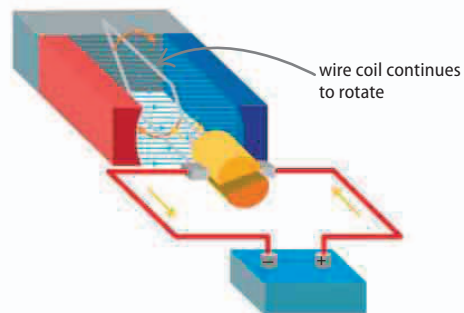
△ Stage 2

Repelled by the permanent magnet's like poles, the wire coil starts turning. After a quarter-turn, the permanent magnets also begin attracting the opposite pole of the wire coil, helping to complete the half-turn.



△ Stage 3

With the poles of the wire coil and permanent magnet now lining up, the commutator reverses the direction of the current in the wire coil. This switches the polarity of the wire coil's magnetic field.



△ Stage 4

With the coil's current reversed, the like poles of the coil and permanent magnet repel again. The coil continues to rotate. When it completes another half-turn, the commutator will reverse the current again to keep the coil spinning.

SEE ALSO

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◀ 203 Electric currents

◀ 210–211 Magnets

Electricity generators

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▽ Left-hand rule

This rule can be used to work out the direction an electric motor turns.

